

## 1 Introduction

### 1.1 Key Results

This report investigates the dynamics of pendulum motion, focusing on the period's dependence on angle and length, and analyzing damping effects through Q-factor measurements.

The motion of the pendulum deviates from the small-angle approximation [1], which assumes a constant period. The period exhibits a **quadratic relationship with the release angle**, expressed as:

$$T(\theta) = T_0(0.0309 \pm 0.005)\theta^2 + \dots,$$

where  $T_0 = 0.795 \pm 0.001$  s at smaller angles, closely matching the approximated value of  $T = 0.79$  s. The **Q-factor** was determined to be  $Q = 85 \pm 8$ , showing a positive correlation with the pendulum's length. For varying string lengths, the period follows the relationship  $T = kL^n$ , with  $k = 1.96 \pm 0.03$  and  $n = 0.50 \pm 0.01$ .

The small-angle approximation is also confirmed when the release angle is small, and the period correlates with length in a manner consistent with the approximation.

The Q-factor of the pendulum is estimated to be  $85 \pm 8$ , with a positive correlation with the pendulum's length. It relates to the decay following  $\theta_0 e^{-t/\tau}$  for some  $\tau$ , further explored in (11) in Section 3.

### 1.2 Small Angle Approximation

Refer to Appendix A. Figure 8 illustrates a pendulum of length  $L$  with a bob of mass  $m$  at  $B$ , relative to the center  $O$ . Initially, the bob is at  $A$ . Let  $\hat{r}$  and  $\hat{n}$  represent the unit vectors along the bob's motion and normal directions, respectively. Assuming:

1.  $L$  is constant with a massless string.
2. Only tension  $F_T$  in the string and gravitational force act on  $B$ .

The forces on  $B$  in  $\hat{r}$  and  $\hat{n}$  directions are:

$$\sum F_B = (|F_T| - mg \cos \theta)\hat{n} + (-mg \sin \theta)\hat{r}. \quad (1)$$

Setting  $s = |\widehat{AB}|$ , we obtain the equation in the  $\hat{r}$ -direction:

$$-mg \sin \theta = m \frac{d^2 s}{dt^2}. \quad (2)$$

Since  $s = L\theta$ , using the small-angle approximation  $\sin \theta \approx \theta$ , we derive:

$$-g\theta = \frac{d^2 s}{dt^2} \Rightarrow \frac{d^2 s}{dt^2} = -\frac{g}{L}s. \quad (3)$$

Thus, under this approximation, the pendulum behaves as a simple harmonic oscillator:

$$s = A \sin \omega t, \quad \frac{d^2 s}{dt^2} = -\omega^2 s, \quad (4)$$

where  $\omega^2 = \frac{g}{L}$  and:

$$T = 2\pi \sqrt{\frac{L}{g}}. \quad (5)$$

### 1.3 Period under Different Lengths

Removing the assumption that length stays constant, we derive a relationship between period and length. A model for the period,  $T$ , associated with the pendulum's length,  $L$ , is given by:

$$T = kL^n, \quad (6)$$

where the model predicts  $k = 2$  and  $n = 0.5$  [1]. This arises from the small-angle approximation (5), where  $T$  is proportional to  $\sqrt{L}$ , and  $2\pi/\sqrt{g} \approx 2$ .

### 1.4 Set Up of the Pendulum

Assuming a positive relationship between the string length and period, the string must be long enough to ensure the period is not too short or difficult to measure.

To accurately assess the angle and period, a clear white background and a small, chromatically sharp bob were chosen to facilitate position tracking.

The setup was optimized for tracker software<sup>1</sup>, which detects specific colors to track the bob's center point.

<sup>1</sup>Developed by Eric Xie, Andy Wen, and I. Available at <https://github.com/HaysonC/PHY180-Pendulum> [2]

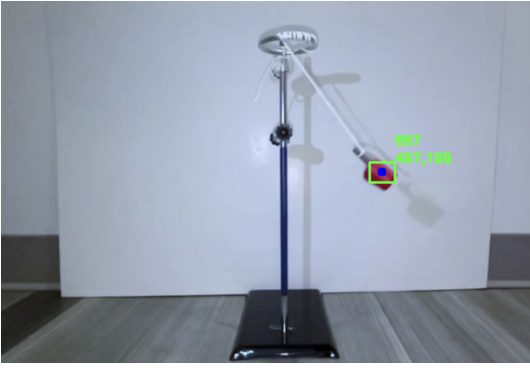


Figure 1: The setup and tracking of the pendulum. The camera captures motion against a white background, using a chromatically sharp bob for precision.

Refer to Figure 1. The experiment used a 1.000 m cotton string secured around a  $70 \pm 0.5$  mm metal ring on a stand. Cotton was chosen for its low elasticity ( $E = 3$  GPa [3]). The string length could be adjusted by wrapping or unwrapping it around the ring, with one end tied to a 50 g steel alloy nut driver (25 mm diameter, 2.6 mm height) as the bob. The string was adjusted to keep the bob off the ground, with a length of  $155 \pm 0.5$  mm measured from the ring to the bob's center of mass. A camera was positioned  $300 \pm 0.5$  mm from the pendulum to capture its motion.

## 2 Period Time and Angles

**Small angle approximation** (5) provides us that period is constant. However, we would like to assess the extent to which this is true for larger angles. This section aims to access the claim that the period is independent of  $\theta$ , and that it follows the formula provided by (5).

**Data Collection** A white backdrop was placed behind the pendulum. The pendulum was to be released starting at more than 60 degrees ( $\pi/4$  radians) to examine the validity of small angle approximation.

**Data Tracking** A tracker software makes use of the stark contrast between the bob's color and the background to determine the location of the pendulum. It records the middle point of the bounding box of the track. The bob is released from 90 degrees ( $\pi/2$  radians) from three times and the recorded footage is fed into the tracker.

### 2.1 Data Processing

Using python, the raw  $x, y$  pixel data is converted to angle,  $\theta$ , data using the following methods:

$$\theta = \arctan \frac{x - x_0}{y - y_0} \quad (7)$$

Where:

$x_0, y_0$  = Center of the pendulum,  
deduced by the max of  $y$  and mean of  $x$

**Period Measurement** In Appendix B, Figure 9 showed the peaks and troughs (anti-peaks) of the pendulum. These are deduced by finding the individual peaks of the data, with some tolerance of noise to prevent multiple or no peaks detected for each oscillation. Thus, the respective time between the peaks is recorded. The positive theta's data series of period is obtained by the peaks and the negative is that of the trough. Period is the time between a peak and the next peak and it would be regarded as the period on that peak.

### 2.2 Uncertainty

**Data Binning** We group the angle data into bins of 0.08 radians. This is chosen as it is the type B uncertainty of most significance, which will be discussed.

It calculates bin edges using the minimum and maximum Angle values, segments the data into these bins and calculates the midpoint of each bin for further analysis. This simplifies the data for easier analysis and visualization.

The above method has some potential uncertainties that need to be addressed.

1. The recorded data is limited to 60 fps, giving a estimated type B uncertainty of  $\frac{1}{2 \cdot 60} = 0.008$ s in the time; and
2. It is certain that the tracker always points within the bob, but the tracker sometimes tracks a point off the center of the bob. Assume that the width of the bob is parallel to its path, we can use half of the width of the bob and the length of the string as two sides of a right triangle. We calculate a value for the uncertainty  $\Delta\theta = \arctan \frac{12.5}{155} = 0.08$ rad. It is  $0.08/\pi = 2.5\%$  of the entire path of pendulum; and
3. The type B percentage uncertainty in the length measurement is  $0.5/155 = 0.3\%$ . Since this is far less than 2.5% it is disregarded; and
4. There are also type A  $u_A(x)$  uncertainties observed in the variations to each data entry on each bin of size  $|X|$ , which is calculated using its standard deviation  $\sigma$  divided by  $|X|$ . This is also used in future calculations of Type A uncertainties.

## 2.3 Analysis

**Small Angle Apprximation** Taking  $L = (0.1550 \pm 0.0005)\text{m}$ ,  $g = 9.806\text{ms}^{-2}$ . We can obtain the period value under the small angle approximation with (5). The percent-age error is propagated with  $L_{\text{err}}/L$

$$T = 2\pi\sqrt{\frac{L}{g}} = 0.7897 \pm 0.003\text{s} \quad (8)$$

**Data** The data obtained, however, noticed a difference from this value, especially at greater values of  $\theta$ .

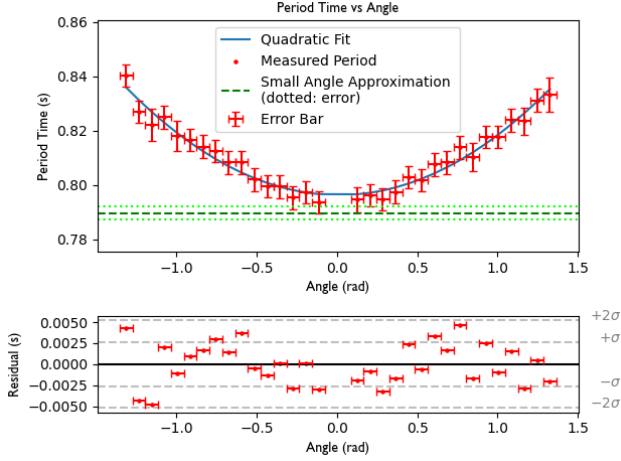


Figure 2: Period Time vs Angle with residue plotted below. The error on angle is half of the bin size and the error on time is the type B or type A uncertainty, whichever is larger.

Referring to figure 2, as seen in the small angle approximation, the period appears constant. However, as we increase the release angle, our data shows a significant deviation from this model, outside of the range of uncertainty, which suggests the need for a more complex, quadratic model [1]. Indeed, the quadratic fit has a high  $r^2 = 0.96$ .

$$T(\theta) = 0.02430 \pm 0.005\theta^2 - 0.0008235 \pm 0.002\theta \quad (9)$$

$$\begin{aligned} &+ 0.7959 \pm 0.001 \\ &= T_0(0.0309 \pm .006\theta^2 - 0.00105 \pm .002\theta + 1.01 \pm .002) \end{aligned} \quad (10)$$

Where:

$$\begin{aligned} T_0 &= [\text{rad}] \text{ Period at the smallest recorded amplitude} \\ &0.79 \pm 0.05 \end{aligned}$$

The coefficients and its variance ( $\sigma^2$ ) are obtained `curve_fit` function in `scipy` [4]. So the uncertainty of the coeeficnet is calculated by dividing  $g \sigma$  with the square-root of the number of data points.

**Agreement at Small Angles** However, we can also see that the small angle uncertainty aligns with the data at a smaller angle. Specifically, we can see that the small angle approximations contain the data points along with the uncertainties.

On the other hand, the quadratic fit at  $\theta = 0$  has the value of 0.7959s. this is outside of the small angle approximations' range of uncertainty.

**Asymmetry** Additionally, asymmetry in the setup, likely caused by the string not being perfectly perpendicular to the camera, introduces further error. Observe, in the residue graph of figure 2, the positive angles' data is generally larger than the right side. Also, 10 shows that the  $\theta$  term is non-zero, suggesting the asymmetry in the fit. However, the difference is less than the range of uncertainty and does not affect the validity of the observations.

While collecting data, we observed that the pendulum spent more time on one side. Upon investigation, we discovered that the location where the cotton string wraps around the metal circle of the stand was not perpendicular to the camera. This caused the swing's plane to be tangent to the point of the circle it wraps around, making it appear as if it was swinging more on one side from the camera's perspective.

**Discussion** Despite some uncertainties and errors in the setup, the data clearly demonstrate that the period is not independent of amplitude, showing a consistent quadratic dependence on angle. At minimal angles, the period aligns with the small-angle approximation within experimental uncertainties.

## 3 Q-Factor and Angle-Time

### 3.1 Exponential Fit

The simple harmonic model assumes no external forces act on the bob, which neglects damping effects such as air drag and string friction. A provided model accounts for air viscosity as the main damping source and assumes a constant period [1]:

$$\theta(t) = \theta_0 e^{-t/\tau} \cos(\omega t + \phi_0). \quad (11)$$

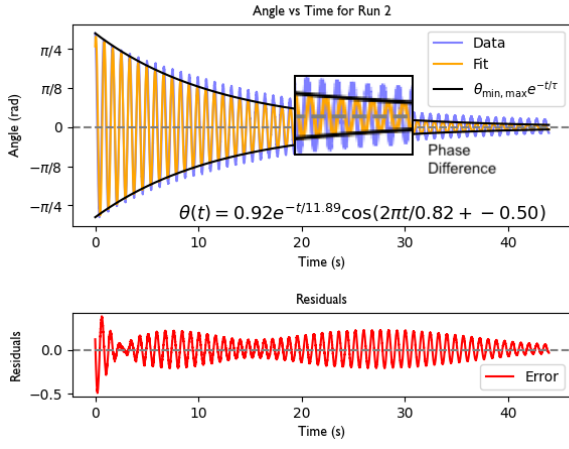


Figure 3: Angle vs. Time for a single trial, showing the exponential decay of amplitude. The curve represents the fit using Equation 11. Deviations highlight discrepancies due to the assumption of a constant period.

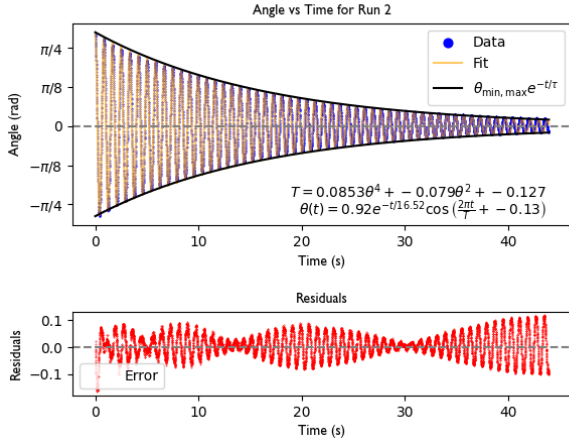


Figure 4: Angle vs. Time for the same trial as Figure 3, with an improved fit incorporating a time-dependent period modeled as a quadratic function of amplitude. The refined fit better aligns with the data, reducing residuals and demonstrating the limitations of simpler models.

Figure 3 shows the best fit for the parameters  $\tau$ ,  $\phi$ , and  $T$ , where  $\theta_0$  is the initial release angle. However, the fit does not align well with the peaks of the data. This discrepancy is likely due to the assumption of a constant period, which does not hold as damping causes the period to vary over time. As the phase difference accumulates, the fit underestimates the actual amplitude, leading to noticeable errors.

To account for this variation, we allow the period to change with time. A suggested approach [5] models the period as a series of even-degree polynomials, simplified to:

$$T_n = a + b\theta_n^2 + c\theta_n^4 + O(\theta_n^6),$$

where  $\theta_n$  is the amplitude at the  $n^{\text{th}}$  period. Since  $\theta_n = \theta_0 e^{-t/\tau}$  represents the maximum amplitude due to decay, the period can be expressed as:

$$T = a + b\theta_0^2 e^{2t/\tau} + c\theta_0^4 e^{4t/\tau} + O(e^{6t/\tau}). \quad (12)$$

Since the series converges, and Section 2 demonstrates that a quadratic fit describes the pendulum's period well, terms up to the quartic term are sufficient for this analysis. Using this improved model, a better fit is achieved, as shown in Figure 4.

The improved fit in Figure 4 achieves  $r^2 = 0.973$ , with significantly smaller residuals compared to Figure 3. This demonstrates that the updated model better represents the pendulum's motion relative to Equation 11.

### 3.2 Q-Factor

**By Period and  $\tau$**  The first method to calculate the  $Q$ -factor, as proposed by [1], uses the ratio:

$$Q = \pi \frac{\tau}{T}. \quad (13)$$

Since  $T$  varies over time, we use the time-averaged period:

$$\bar{T} = \frac{\int_0^{t_{\max}} T dt}{t_{\max}}.$$

The  $Q$ -factor, averaged over three trials, accounts for both Type A (random) and Type B (systematic) uncertainties. This method yields:

$$Q = \pi \frac{\tau}{\bar{T}} = 86 \pm 8. \quad (14)$$

**By Counting Periods** The  $Q$ -factor can also be defined as the number of oscillations required for the amplitude to decay to  $e^{-\pi} \approx 4\%$  of the initial amplitude [1]. Since the data do not extend to this point, we use  $e^{-\pi/4}$  as an intermediate value for  $Q/4$  and count the corresponding number of periods. Using uncertainty propagation, this method gives:

$$Q = 4 \cdot |\{T_n | \theta_n > \theta_0 e^{-\pi/4}\}| = 84 \pm 8. \quad (15)$$

**Mean Q-Factor** Averaging the results from both methods gives:

$$Q = \frac{86 + 84}{2} = 85 \pm 8. \quad (16)$$

The  $Q$ -factor values from both methods are consistent within the margin of uncertainty, providing strong evidence that the pendulum's  $Q$ -factor lies in the range 70–90, depending on experimental conditions and damping factors.

## 4 Period; Q-Factor and Pendulum Length

**Data Collection** To examine the relationship between length,  $Q$  and period, we record the pendulum starting from a fixed length. By wrapping the string around the ring, we can easily adjust its length as needed. Once to obtain the length of the next trial. Hence, the lengths recorded are, in mm, [113, 135, 165, 185, 213, 239]

We start from  $10 \pm 0.5$  degrees and record the rate of decay as the for  $Q$  and also period time. The recording stops after 20 seconds.

By releasing it at a small angle, the period and  $Q$  factor would be minimally affected by the variation in period demonstrated by section 2, especially when it is only varying less than five percent.

### 4.1 Period and Length

**Line of Fit** A model of period and length is given by (6). For this, we use a line of best fit that varies most to  $k$ , and  $n$  values, we obtained a figure of  $k = 1.96 \pm 0.03$  and  $n = 0.50 \pm 0.01$  with a very high  $r^2$  value of 0.998.

A plot of the data collected is presented below, with the length error being  $\pm 0.5$  mm, which is too small to show.

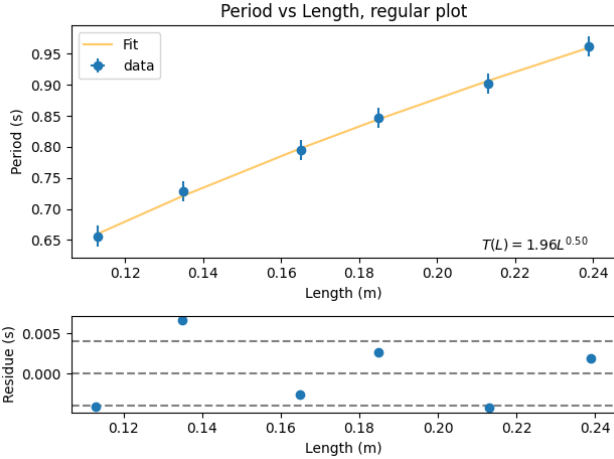


Figure 5: Period vs. Length in linear scale for all trials. The data show a clear trend consistent with the model  $T = kL^n$ , where the fit yields  $k = 1.96 \pm 0.03$  and  $n = 0.50 \pm 0.01$ , demonstrating the square-root relationship between period and length.

And below is the same plot in a log-log scale.

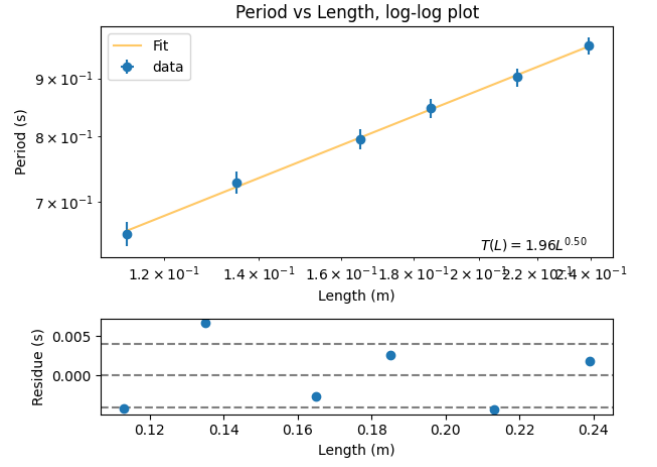


Figure 6: Period vs Length graph in log-log scale for all trials, with the proposed line of best fit

The square-root relationship between length and period is apparent as the data follow the straight line in Figure 6 closely, resulting in a high  $r^2$  value.

Though random type B uncertainty of 10% in the release angle influenced the period's data as period is demonstrated to be not constant against release angle, it seems to be less than that of the limitation of the recording's fps, which is the sole contributor to the time uncertainty.

The data also systematic error. This is clear since there is a high  $r^2$  value to a fit, but has  $k = 1.93 \pm 0.03$ . While the model predicts a value of 2, and  $2\pi/g$  have a value slightly over 2,  $k$  is below that figure outside of the margin of the fit's uncertainty.

It is quite certain that it is not due to the small angle approximation as it was demonstrated that a high release angle would only increase the period and hence increase  $k$ .

This demonstrates that the models hadn't accounted for some marginal error factors that the setup has introduced. These could be friction or damping. Hence, this indicates opportunities for improvements by the models by adding factors to its differential equation.

### 4.2 Q factor and length

Following the period analysis, we now examine the role of the  $Q$ -factor in damping and its dependency on length. To determine  $Q$  factor, we use the equation from (14) in section 3. This is done by using the same exponential fit in section 3 to obtain the  $\tau$  and  $\bar{T}$  values.



However, the relationship between length and  $Q$ -value seems to be less conclusive. Below is a linear fit of the  $Q$  value and length.

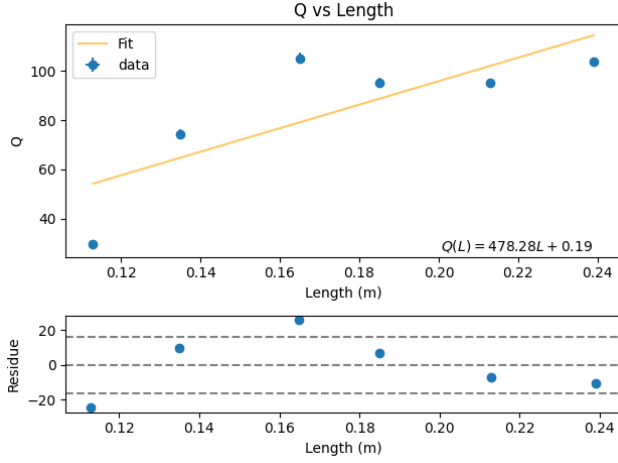


Figure 7:  $Q$ -factor vs. Length with a linear fit. While an increasing trend in  $Q$ -factor with length is observed, the data show significant scatter, resulting in a low  $r^2$  value of 0.62. This suggests that random and systematic errors in damping measurements may obscure a more precise relationship.

The linear fit models that the  $Q$ -value follows a relationship of  $Q = (500 \pm 200)L + 0 \pm 30$ . A linear model was chosen for simplicity, as higher-order fits introduced overfitting with no clear physical interpretation.

Type B uncertainty in length measurement contributes most to overall length uncertainty, which is  $\pm 0.0005$  m. The type A uncertainty of the  $Q$  value can't be determined as section 3, since it is obtained in one trial instead of multiple trials. Thus, the major uncertainty of  $Q$ -value is obtained via propagating the relative uncertainty of the  $\tau$  value from the exponential fit, using the same method in section 3.

Although there seems to be a general (increasing) correlation there-between, the data does not follow closely to the fit and it has a low correlation coefficient of  $r^2 = 0.62$ .

The constant coefficient for the fit could be treated as experimentally zero within the margin of error. If the linear model is correct, the  $Q$  value is directly proportional to length. This confirms the model provided in the lecture [1], which indicates the  $Q$  value proportional to length at smaller length values.

There is also a pattern seen in the residence graph. However, it is believed that it is just a coincidence as there is not significant support for other types of fit, as higher-order functions lead to overfitting.

**Discussion** While there is significant variation between the fit and the data, an increasing trend of  $Q$  value is observed which agrees with the observation made by the lecture. Taking from the same set of data, however, it is believed that the reason for such a low value of  $r$  for the  $Q$ -factor, compared to that of the period, could be the following:

1. Period depends mainly on the length and  $g$  which variations is none, while the decay constant,  $\tau$ , would be significantly impacted by highly random, immeasurable errors such as wind and variations.
2.  $Q$ -factor's calculation involves many steps, including fitting the decay of the data which involves different kinds of type A and type B uncertainties that are unaccounted for. These uncertainties accumulate.

## 5 Conclusion

**Period and Small Angle Approximation** It is believed that the period of the data varies against angle quadratically (or at least varies in a parabolic format). The experiment showed that the period is at minima when the angle is small, equal to the small angle approximation given the uncertainties.

The period versus lengths experiment demonstrated that the small angle approximation is accurate to some extent when the length is varied, while it could be improved to incorporate elements of damping in its differential equation.

**$Q$ -factor** The  $Q$ -factor was determined to be approximately  $85 \pm 8$ , with an observed increasing trend with length.

### 5.1 Uncertainities

Type A uncertainty, due to variability across trials, contributes approximately less than 10% to  $Q$ -factor, and less than 5% of period time, notably affecting higher angle value. Type B uncertainties, arising from equipment constraints such as the 60 fps frame rate and the  $\pm 0.5$  mm length measurement tolerance, affect individual reading accuracy by an estimated 1–2%. Together, these uncertainties are within an acceptable range ( $\pm 5$ –7%), reinforcing the reliability of the overall trends and conclusions.

**Time** These period data and  $Q$  data showed errors affecting the data, particularly in time, although it only accounts for a relatively small part of the around 0.8 readings. Since the period time is short, the frame rate (60 fps) of the recording becomes a relatively significant factor of uncertainty, as the type B uncertainty could be estimated to

$1/60/2 = 0.008s$ , which is 20% the variation in a period of section 2.

**Varying Length** The type A uncertainty seems to be small with a high  $r^2$  value suggesting low statistic variance for the period and length experiment. When the length of the pendulum is varied, there are also random variations of the data leading to uncertainty in the model. However, a systematic error suggested a flaw in the model for [1].

## 5.2 Improvements

**Multiple Trials** When length is varied, to improve the quality of the data, it would be best to conduct multiple trials per length, to eliminate random error.

**Pendulum Setup** Moreover, work could be done regarding the pendulum setup. The ‘wrapping around’ method causes significant variation in fraction while the pendulum oscillates since the string rests on different surfaces between the cotton string itself the metal ring, or both at the same time. We could instead tie it on a round hook and adjust its length by adjusting the tie.

**Recording Time** The equipment shall also be improved to record higher fps data, to decrease the uncertainty and random error caused by time intervals, and to obtain a better result of period time.

**High Release Angle** In addition, as section 2 concluded that period time is not constant over different release angles, the release angle of 10 degree is relatively small to the pendulum’s range of possible motion. To further examine the validity of the approximation proposed by [1] when the length is varied, we could have a larger release angle further to the domain of a small angle approximation.

## References

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- [2] H. Cheung, Y. Wen, and E. Xie, “Phy180-pendulum,” 2024. Available Online at: <https://github.com/HaysonC/PHY180-Pendulum>.
- [3] Jelgava, G. Kharkova, O. Kononova, A. Krasnikovs, M. Eiduks, E. Machanovskis, and K. Dzelzitis, *ENGINEERING FOR RURAL DEVELOPMENT ELASTIC PROPERTIES OF COTTON FABRIC BASED POLYMER COMPOSITES*. 2011.
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- [5] C. U. of Technology, “The pendulum - rich physics from a simple system,” 1985. Accessed: 2024-10-04.



## A Free Body Diagram of a Penulum

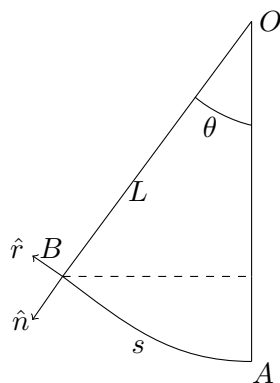


Figure 8: A Free Body Diagram of a Pendulum

## B Peaks and Troughs of a Pendulum

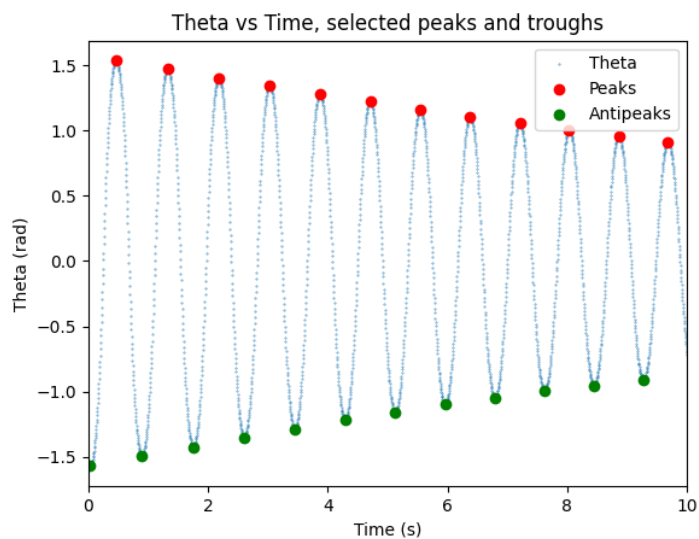


Figure 9: The time between the troughs and peaks of the data could be thought of as the period for a specific angle